Database Systems Laboratory

Course No: CSE-3110

Database Project Report



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Title: Pharmacy Management System

Project Description:

The Pharmacy Management System (PMS) is a comprehensive software solution designed to automate and streamline the day-to-day operations of pharmacies and medical stores. The system focuses on improving operational efficiency, maintaining accurate records of medicines, purchases, sales, inventory, and transactions, while ensuring that pharmacists and store managers can manage their operations in a hassle-free, accurate, and organized manner.

Modern pharmacies face challenges such as manual record-keeping, inventory mismanagement, errors in medicine dispensation, tracking expiry dates, and handling complex sales and returns processes. This system addresses these challenges by providing an integrated platform that handles all critical aspects of pharmacy management.

The Pharmacy Management System is developed using a robust database design, ensuring that all entities, relationships, and operations are accurately modeled. The system supports functionalities such as managing medicine types, medicines, suppliers, purchase and sales operations, stock tracking, returns management, and financial transactions. It also includes user management for different roles, such as admin, pharmacist, and cashier, ensuring proper authorizartion and control over sensitive operations.

Discussion:

The database in this Pharmacy Management System acts as the central backbone of the entire application, ensuring smooth data flow between various functional modules such as User Management, Inventory, Sales, Purchases, and Transactions.
Each module is interconnected through primary and foreign key relationships, ensuring data integrity and accountability.

1.User Management:

-The Users table stores login credentials, personal details, and roles (e.g., Admin, Pharmacist, Cashier).
-Each user is assigned a unique ID, and roles define access permissions within the system.

-The system maintains secure passwords (hashed) and records all user activities — such as who created a sale, who entered purchase data, etc.

-This ensures accountability — every change or transaction in the system can be traced back to the responsible user.

2.Medicine and Inventory Management

1.The Medicines table contains detailed information for each product:
SKU, Generic Name, Unit, Strength, Price, and Reorder Level.

2.Each medicine can have multiple batches, stored in the Batches table, each identified by a batch number, expiry date, quantity, and cost.

3.When purchases or sales occur, the batch quantities are automatically updated.

4.If the stock level of any medicine drops below its reorder threshold, the system highlights it as low-stock for timely replenishment.

3.Supplier and Purchase Management:

1.The Suppliers table keeps information about all suppliers, including name, contact number, and address.

2.When a pharmacy buys medicines, a Purchase Order (PO) is created in the Purchases table, linked to a supplier and the user who entered it.

3.Each purchase may include multiple medicines stored in a Purchase Items table, with quantities, costs, and batch details.

4.Once a purchase is recorded, the inventory automatically updates stock levels and creates new batch entries.

4.Sales and Customer Management:

-Each sale is represented by a Sales Invoice in the Sales table, containing details like invoice number, date, total amount, tax, and discount.

-Sales Items table stores all medicines sold per invoice, referencing both the medicine and batch to ensure correct stock deductions.

-The Customer table keeps optional customer details (for regular or credited sales).

-The Sales table also links to the user (cashier/pharmacist) who handled the sale, providing transparency in operations.

5.Returns and Transactions Management:

-When medicines are returned (by customers or to suppliers), a Returns table records the transaction details, refund amounts, and associated sale or purchase.

-The Transactions table logs all payments, whether they are for sales, returns, or purchases.

-Each transaction includes amount, payment method (cash, card, etc.), and user ID to identify who processed the payment.

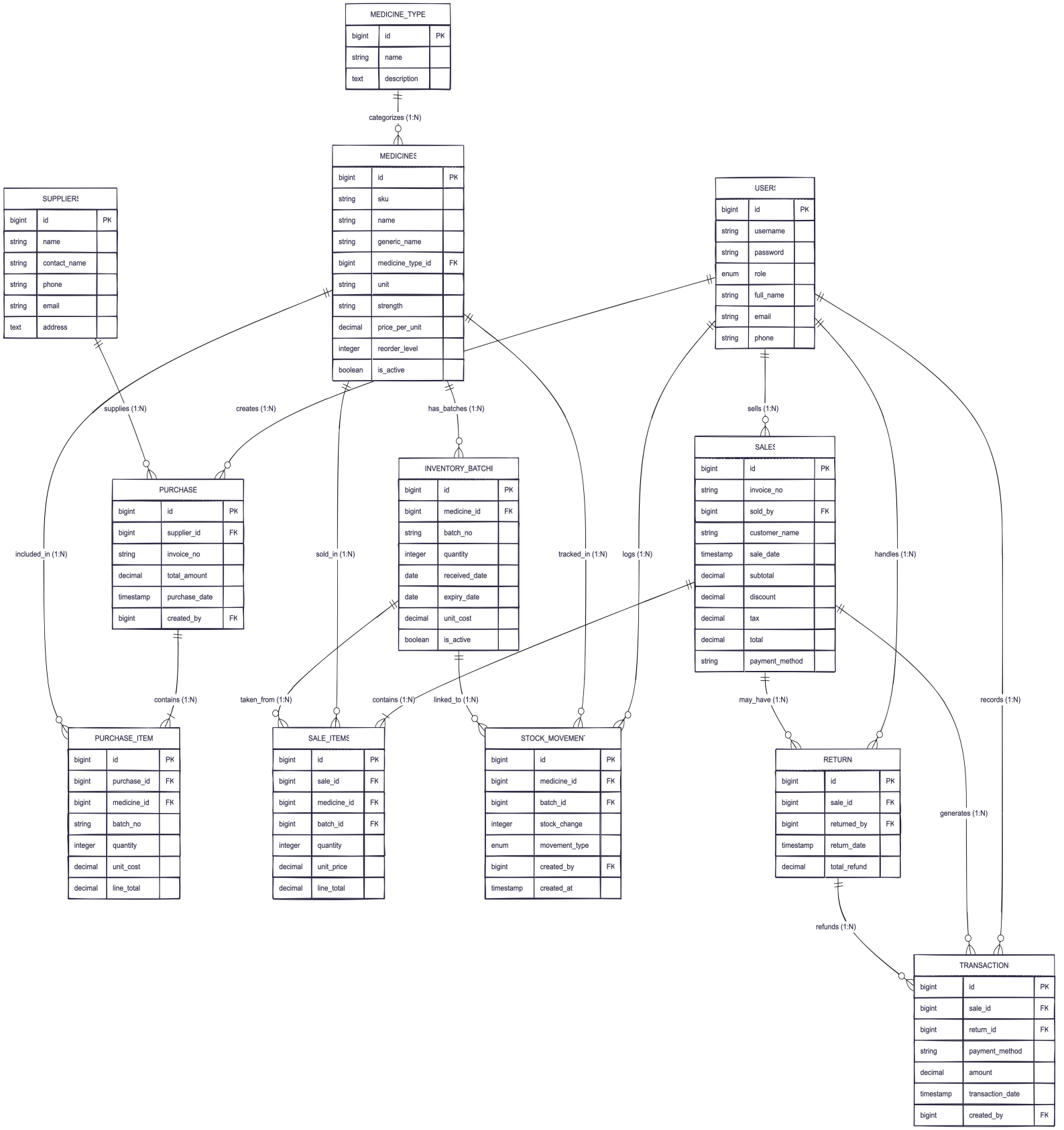
6.Stock Movements Tracking:

-The Stock Movements table keeps a detailed audit of all inventory changes:

- Purchases (Incoming)
- Sales (Outgoing)
- Returns (Adjustments)
- Manual corrections (if any)

- Each movement is timestamped and linked to the responsible user, ensuring traceability.
- This is crucial for auditing and maintaining inventory accuracy.

Schema Diagram:



Entity Relationship(ER) Diagram:

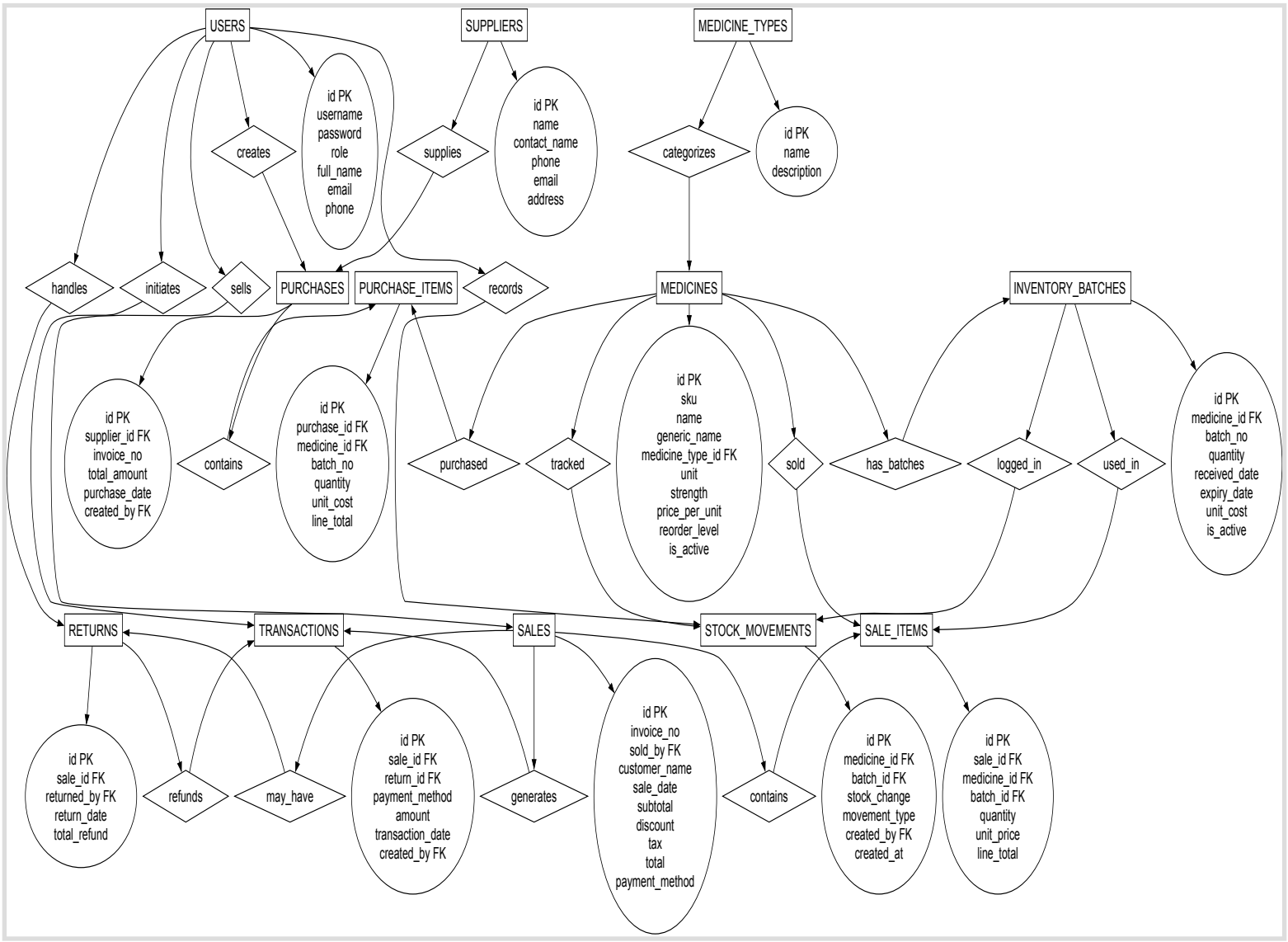


Fig 2 : ER diagram of Pharmacy Management

Conclusion:

The Pharmacy Management System is a complete solution for pharmacies that want to automate their operations, maintain accurate data, improve efficiency, and provide better service to their customers. By integrating user management, medicine tracking, sales, purchases, returns, and transactions into a single platform, the system ensures that all pharmacy operations are transparent, auditable, and easy to manage.

With this system, pharmacies can save time, reduce errors, improve inventory control, and make informed business decisions. Its robust ER database design ensures data integrity and supports future scalability for advanced features like analytics, automated reorder alerts, and integration with online sales platforms.